

Reducing Crime Without the Police?

Ignacio Sarmiento-Barbieri¹ Michael Weintraub¹
Carlos Díaz² Olga Lucía Sarmiento¹

¹Universidad de los Andes (Colombia)

²Universidad Alberto Hurtado (Chile)

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Motivation

Crime is a challenge in developing countries, yet policing alone often delivers limited results

- Police capacity, incentives, and oversight are often weak
- Enforcement-based strategies have uneven effects (Collazos et al. 2021; Blattman et al. 2023)
- This motivates interest in complementary approaches

Can changes to the built environment improve public safety?

- Small-scale interventions (street lighting, tree cover) show promise (Chalfin et al. 2022; Branas et al. 2016)
- But evidence on large-scale transport infrastructure is limited
- And crime effects are rarely incorporated in welfare evaluation of transport projects

Motivation

"Having MARTA [rail] increases the chances of crime. The criminals can get off the train and break into your home and get back onto the train. I don't think that's as likely to happen on buses, where they have to walk past the driver to get on board."

Source: Atlanta Journal Constitution (1997)

Motivation

 **Joe Kent for Congress WA-03**
1d · 🌐

Even though Clark County has rejected light rail three times, [Marie Gluesenkamp Perez for Congress](#) supports using federal dollars for a new light rail project that will bring Portland's crime, homelessness, and drugs to Clark County and beyond.

Say NO the Marie Perez Express!

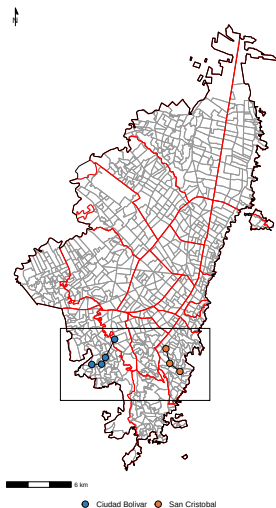


Source: Bloomberg (2023)

This paper

We study *TransMiCable*, a gondola lift system in Bogotá, Colombia.

- Built in Ciudad Bolívar: homicide rate 42/100k, $\sim 1.7\times$ city average
- Inaugurated in 2018; 3.4 km, three stations
- Large, highly visible public investment (USD\$109M)
- Connects to TransMilenio BRT (largest system globally)



TransMiCable in Bogotá



Theoretical framework (intuition)

Infrastructure projects reshape the spatial equilibrium of crime

- Offenders choose *where* to operate, with payoffs that depend on:
 - monitoring and detection risk
 - target density and value
 - costs of travel and descent from the hillside
- TransMiCable changes multiple margins simultaneously:
 - ↑ activity and informal monitoring near stations (deters local crime)
 - ↑ legal wages from construction employment (raises opportunity cost)
 - ↓ travel costs to downstream nodes (may shift crime along network)
- Net effect is theoretically ambiguous; we take it to the data

Testable predictions

- **Local deterrence (near stations)**
 - Crime falls near stations during construction *and* after opening
 - Effects decay sharply with distance; no broad locality-wide change
 - Construction-phase reductions precede passenger flows
- **Network displacement**
 - Improved access lowers cost of reaching downstream nodes
 - Opportunistic street theft may rise at connected transit stops
- **Crime-type specificity**
 - Street-level thefts most likely to displace
 - Crimes tied to private spaces (burglary, domestic violence) should not
- **Net effect**
 - Station-area reductions > feeder-stop increases $\Rightarrow$ corridor-wide safety gain

Data and identification

Data

- Geo-coded administrative crime data from the National Police, 2011–2020
- Quarterly panel: 100m × 100m grid cells

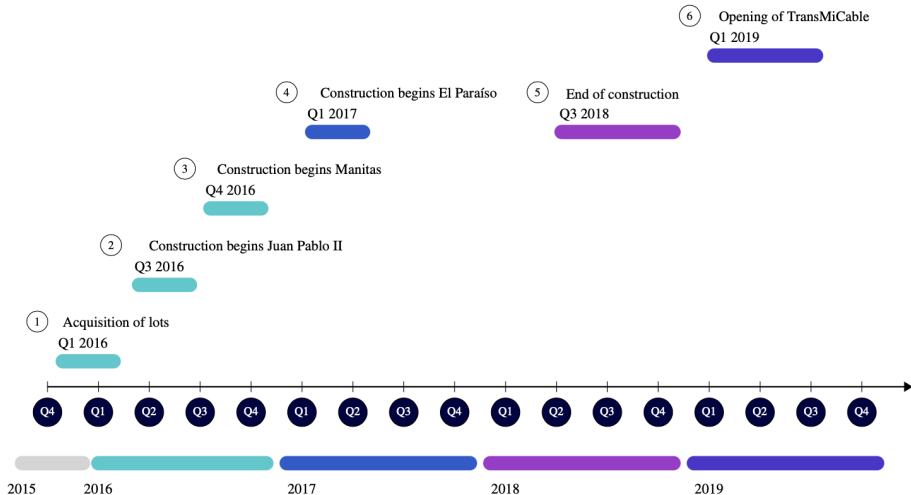
Identification: spatial difference-in-differences

- Compare grid cells $\leq 800\text{m}$ of stations to cells 800m–2km away
- All comparisons *within* Ciudad Bolívar
- Grid-cell FE + quarter-year FE; SEs clustered by neighborhood
- 800m threshold: motivated by observed spatial decay in treatment effects

Two treatments, two estimators

- **Construction start** (staggered: Q3 2016–Q1 2017) → Gardner (2022) staggered DiD
- **System opening** (common: Q4 2018) → standard TWFE event study

Timeline: construction and opening



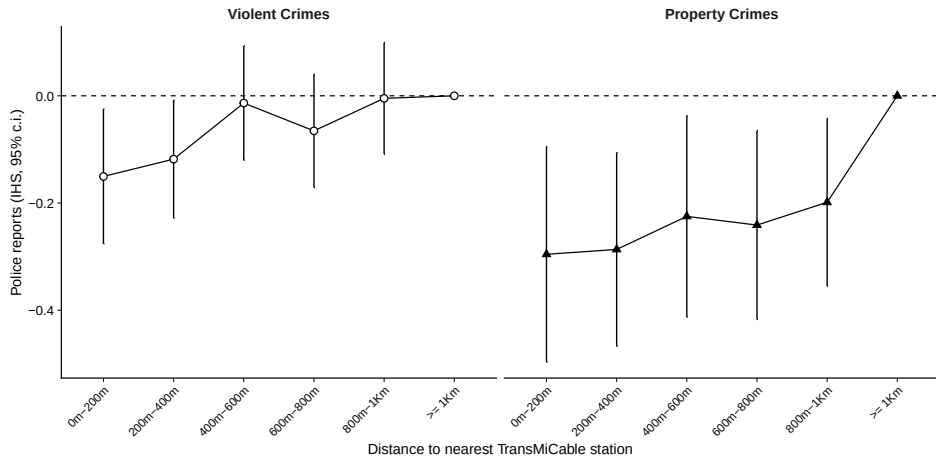
Main results: crime falls near stations once operation begins

Significant reductions in crime within 800m of stations

	Total Crime (IHS)	Violent Crime (IHS)	Property Crime (IHS)
1[Distance $\leq$ 800m] $\times$ Post Opening	-0.1649*** (0.0476)	-0.0717** (0.0287)	-0.1365** (0.0616)
Implied semi-elasticity	$\sim -16.5\%$	$\sim -7.2\%$	$\sim -13.7\%$
Grid-cell FE / Quarter-year FE	Yes	Yes	Yes
Observations	296,064	296,064	296,064

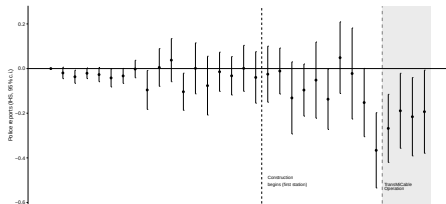
Property crime drives the catchment effect. Effects robust to OLS-in-levels, Poisson, and negative-binomial.

Crime reductions are highly localized

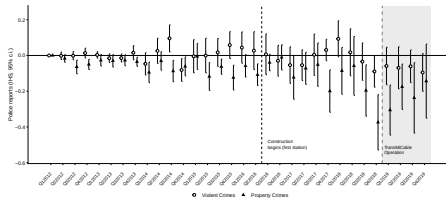


Violent crime decays by ~400m; property crime decays gradually through 800m. Motivates 800m catchment cutoff.

Event study: sharp crime decreases after opening, no pre-trends

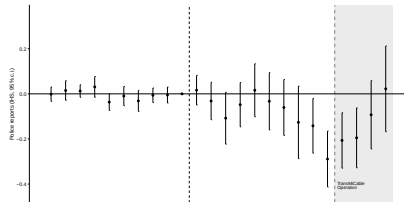


(a) Total crime

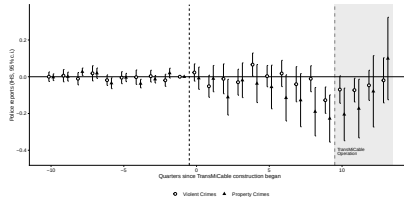


(b) Violent and property crime

Crime begins falling *during construction*, before a single rider

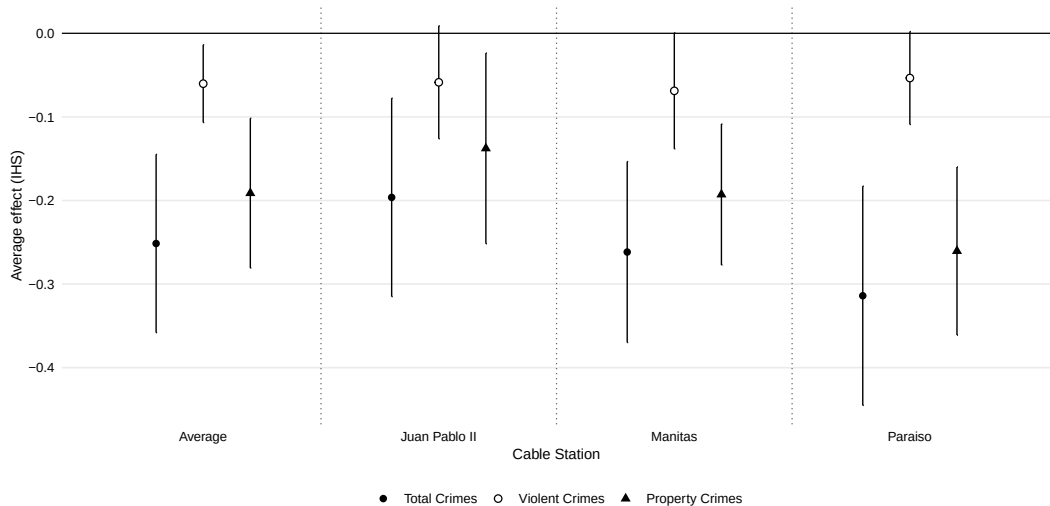


(a) Total crime

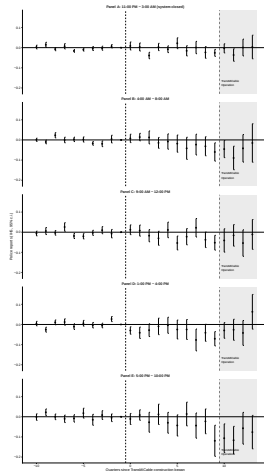


(b) Violent and property crime

Consistent effects across all three stations



Mechanism check: effects track passenger flows by time of day



B–E (open): decline, strongest 5–10 PM. **A** (closed): attenuated $\Rightarrow$ built-environment.

But did crime simply displace across space?

Place-based interventions may reallocate crime throughout the public transit network.

Theory predicts relocation toward nodes with:

- higher target density
- greater anonymity
- lower travel costs

We examine two downstream locations:

- Terminal node (Portal Tunal)
- Feeder bus stops (SITP componente zonal)

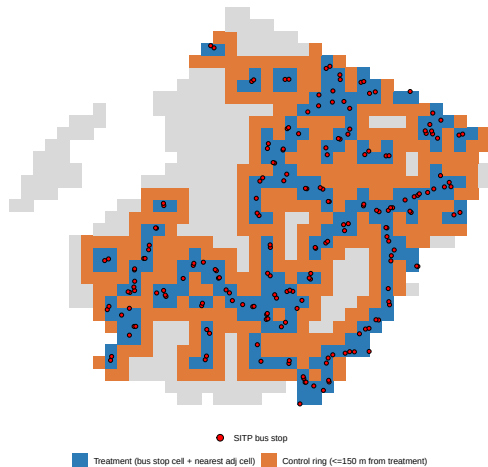
Displacement at the terminal node (Portal Tunal)

	Total	Violent	Property
Intermediate stations $\leq 800\text{m} \times$ Construction	-0.094***	-0.014	-0.107***
Intermediate stations $\leq 800\text{m} \times$ Post Opening	-0.208***	-0.065***	-0.207***
Tunal $\leq 200\text{m} \times$ Construction	+0.108***	-0.207***	+0.291***
Tunal 200–400m $\times$ Construction	-0.104**	-0.119***	-0.023
Tunal 400–600m $\times$ Construction	-0.155***	-0.202***	+0.004
<i>Tunal $\leq 200\text{m} \times$ Post Opening</i>	+0.065	-0.330***	+0.307*
Tunal 200–400m $\times$ Post Opening	-0.229**	-0.149***	-0.125

Within 200m of Tunal: **property crime rises**, violent crime falls sharply. Crowding at the BRT transfer creates theft opportunities while still deterring violence. Effects non-monotonic and highly localized.

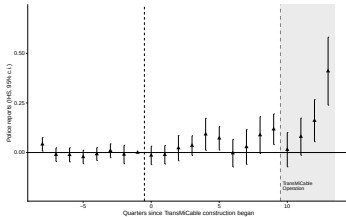
Does crime spill over to the feeder network?

- SITP feeder stops concentrate pedestrians but lack formal surveillance: natural displacement margin
- Treatment: cells with an SITP stop (+nearest adjacent cell)
- Control: non-stop cells within 150m
- Staggered timing inherited from nearest TransMiCable station
- Gardner (2022) two-stage estimator

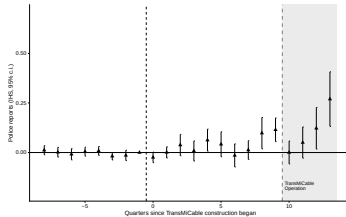


Displacement is offense-specific: street theft up, violence flat

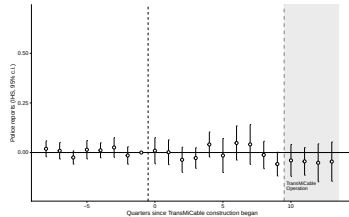
Thefts from persons ↑



Cell phone thefts ↑

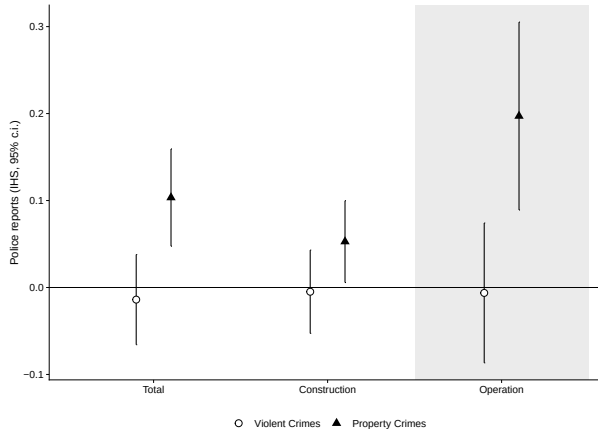


Domestic violence (placebo) ×



Motor vehicle thefts and burglaries also show no effect. Domestic-violence placebo rules out a generic locality-wide trend.

Cumulative feeder effect: property crime up, violent flat



Property crime rises post-opening at feeder stops; violent crime is flat throughout.

Net welfare: is the corridor safer overall?

Margin	$\hat{\beta}$ (per cell, qtr.)	N cells treated	Quarterly Δ (crimes)	Total Δ (crimes)
Catchment (3 hillside stations)	-0.502***	6,898	-3,461	-13,842
Tunal terminal, 0-200m	+1.209*	39	+47	+189
Tunal terminal, 200-400m	+0.101	190	+19	+77
Tunal terminal, 400-600m	-0.062	439	-27	-109
Tunal terminal, 600-800m	+0.141	793	+112	+447
Feeder bus stops (SITP)	+0.250**	213	+53	+213
Net corridor effect				-13,026

Spatial displacement is real but **second-order**: Tunal + feeder spillovers attenuate the catchment gain by only $\approx 6\%$. Quality margin: displacement shifts *property* offenses $\rightarrow$ feeder stops, while *violent* crime continues to fall corridor-wide.

Conclusion

Large-scale infrastructure projects reshape *where* and *which* crimes occur

- TransMiCable cuts crime within 800m of stations by ~**16.5%** (driven by ~**13.7%** property; ~**7.2%** violent): **no policing required**
- Effects begin *during construction*, before a single rider: monitoring precedes commuter flows
- Within-day signature tracks operating hours $\Rightarrow$ embedded place management
- Displacement is offense-specific: opportunistic theft relocates to feeder stops; motor-vehicle theft, burglary, intra-household violence do not
- Net corridor reduction $\approx$ **13,026** property crimes (post-opening, 4 quarters); only $\approx$ 6% attenuation
- Safety co-benefits belong in welfare evaluation of transport investments; the unit is the *network*, not the catchment

Thanks!

Ignacio Sarmiento-Barbieri
Universidad de los Andes

i.sarmiento@uniandes.edu.co
<https://ignaciomsarmiento.github.io/>

Change in how crime is measured

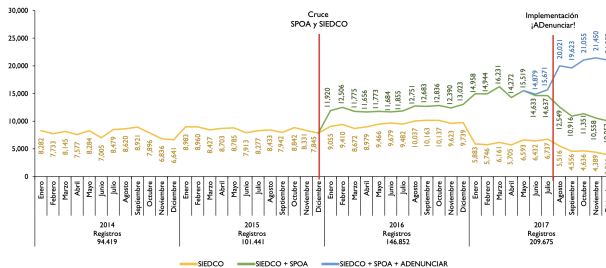
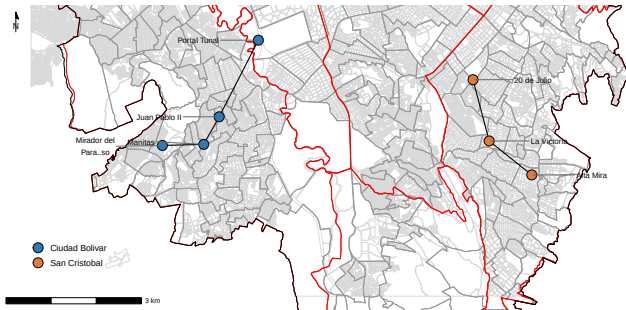


Figura 7. Cantidad de hurtos a personas según fuente de la información, Colombia, 2014-2017 (mensual)

Fuente: Policía Nacional (2018). Elaboración propia.

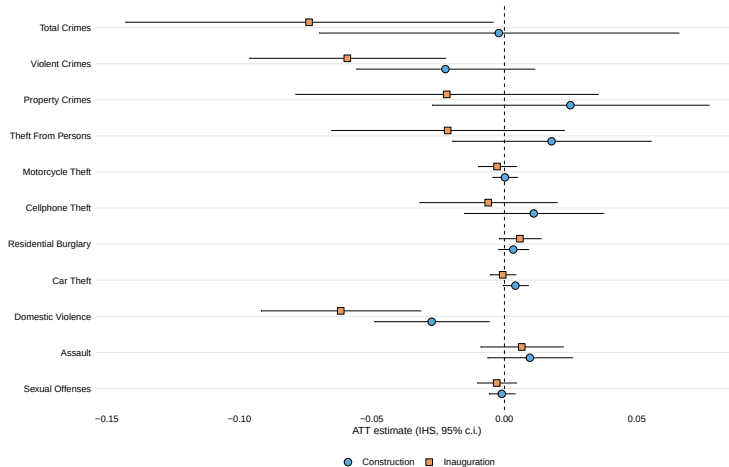
Reproduced from Rodríguez, J. D., Mejía, D., Caro, L., Romero, M. & Campos, F. (2018). Implicaciones del proceso de integración de los registros administrativos de criminalidad entre el SPOA de la Fiscalía General y el SIEDCO de la Policía Nacional de Colombia, y la puesta en marcha del aplicativo “¡ADenunciar!” sobre las cifras de criminalidad. *Revista Criminalidad*, 60(3): 9–27.

Robustness: triple differences with San Cristóbal (future sites)

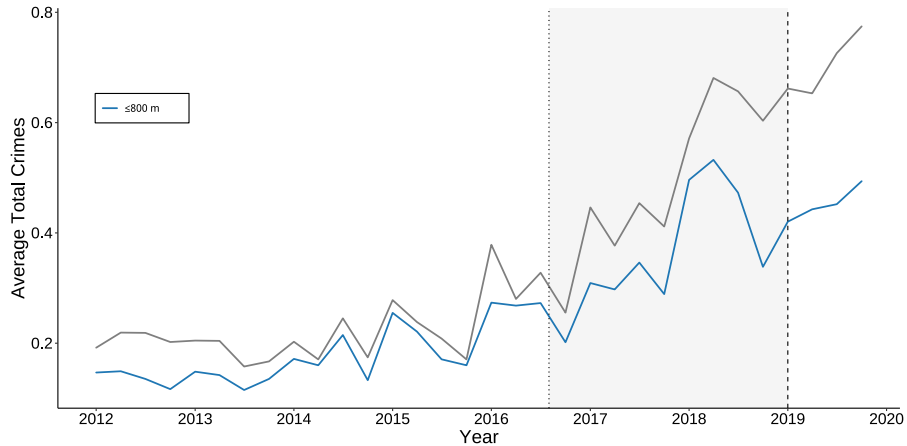


San Cristóbal: similar peripheral area designated for future TransMiCable, no construction during the study period; high crime and low transit access make it a credible counterfactual.

Robustness: triple differences with San Cristóbal (future sites)

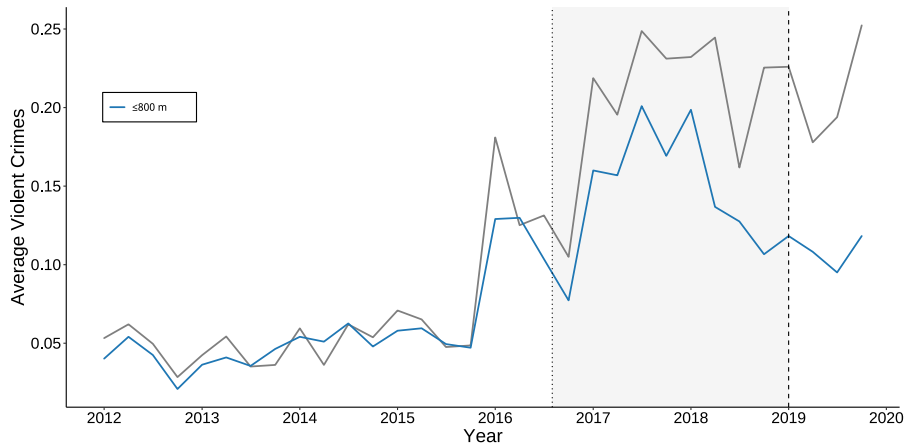


Raw near vs. far trends: Ciudad Bolívar



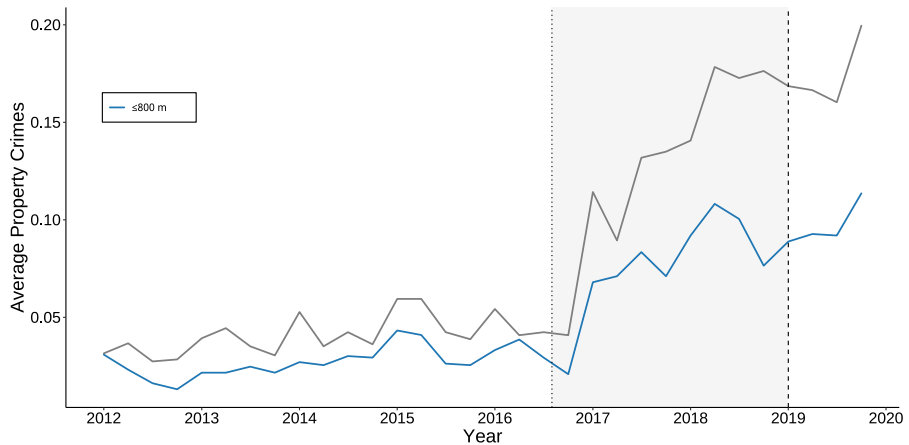
(1/3) Total crime

Raw near vs. far trends: Ciudad Bolívar



(2/3) Violent crime

Raw near vs. far trends: Ciudad Bolívar



(3/3) Property crime